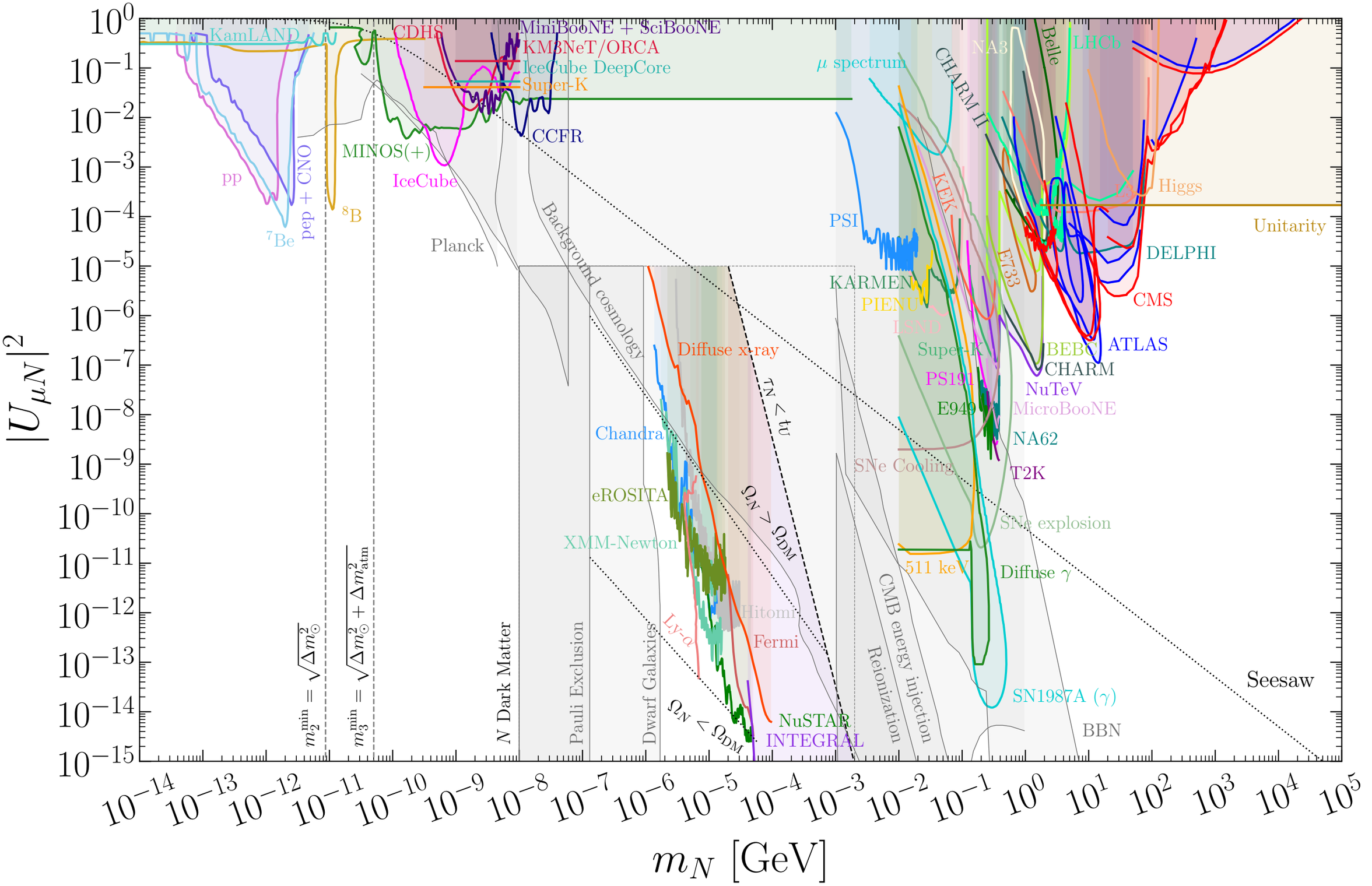

$|U_{\mu N}|^2$
$m_N$ [GeV]
KamLAND
pp
$^7$Be
pep + CNO
$^8$B
CDHS
MINOS(+)
IceCube
MiniBooNE + SciBooNE
KM3NeT/ORCA
IceCube DeepCore
Super-K
CCFR
Planck
Background cosmology
$\mu$ spectrum
CHARM II
NA3
Belle
LHCb
PSI
KEK
E733
Higgs
Unitarity
DELPHI
KARMEN
PIENU
LSND
CMS
Super-K
ATLAS
BEBC
CHARM
PS191
NuTeV
E949
MicroBooNE
NA62
T2K
SNe Cooling
SNe explosion
511 keV
Diffuse $\gamma$
SN1987A ($\gamma$)
BBN
Seesaw
Diffuse x-ray
$\tau_N < t_U$
$\Omega_N > \Omega_{DM}$
Chandra
eROSITA
XMM-Newton
Ly-$\alpha$
Hitomi
Fermi
NuSTAR
INTEGRAL
$\Omega_N < \Omega_{DM}$
$N$ Dark Matter
Pauli Exclusion
Dwarf Galaxies
CMB energy injection
Reionization
$m_2^{\rm min} = \sqrt{\Delta m^2_\odot}$
$m_3^{\rm min} = \sqrt{\Delta m^2_\odot + \Delta m^2_{\rm atm}}$